

# Teaching Tech Together

## Cognitive Capacity



Greg Wilson  
<https://third-bit.com>



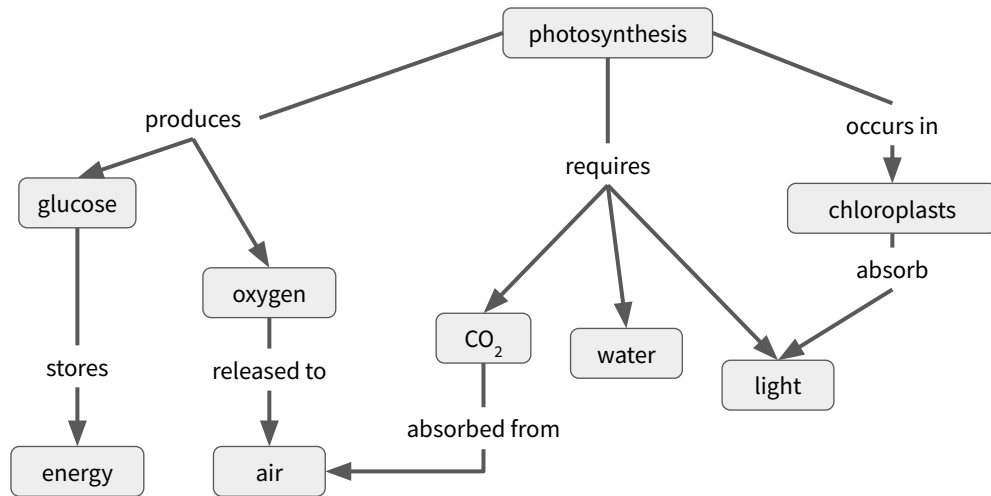
Estimated time: 40 minutes.

All art from [openclipart.org](https://openclipart.org) (open license)

### Learning Objectives

1. Explain the “magic number  $7 \pm 2$ ” (capacity of short-term memory) and its implications for lesson design and delivery.
2. Explain what a faded example is and how its effectiveness can be grounded in the limited capacity of short-term memory.
3. Create faded examples and explain when and how they should or shouldn’t be used.

## Concept Maps Revisited

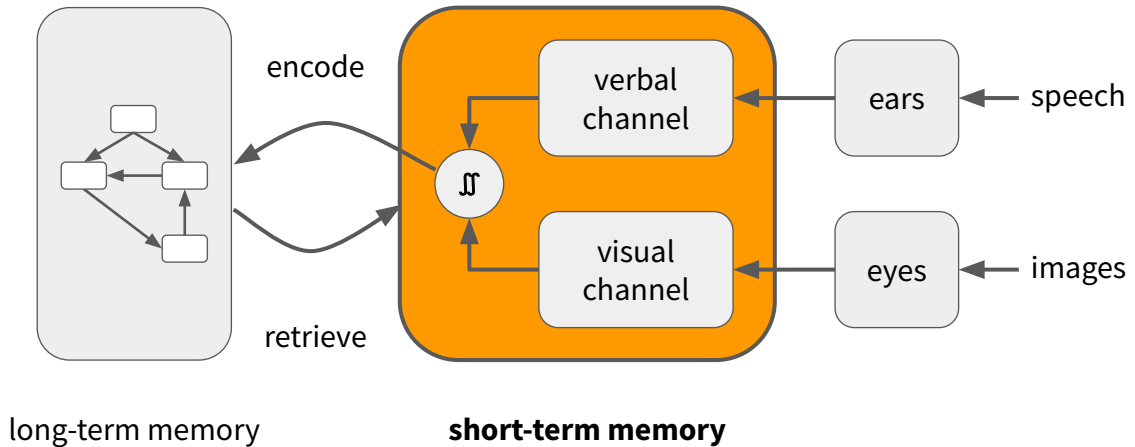


2

We said earlier that instructors should divide their overall concept map into pieces with formative assessments after each piece. But how? And how big should the pieces be?

Description: concept map of photosynthesis including nine concepts, which are the nodes in the graph, and seven labeled arcs, some of which are one-to-many.

# Cognitive Architecture (Simplified)

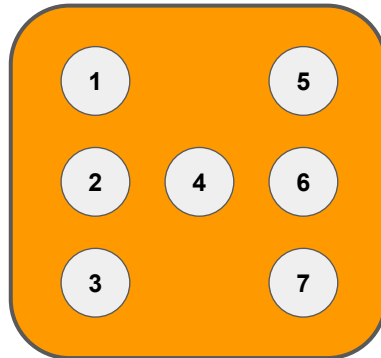


3

To answer that question, let's come back to our cognitive architecture diagram and take another look at short-term memory.

Description: the cognitive architecture diagram from the previous slide deck with the central box representing short-term memory emphasized. The box on the left is labeled "long-term memory" and contains a graph representing a concept map. It is connected to the central box representing short-term memory with arcs labeled "retrieve" and "encode". The short-term memory box contains smaller boxes labeled "verbal channel", "visual channel", and an integrals sign, which represents the integration of verbal and visual information. To the right of the short-term memory box are two smaller boxes labeled "ears" and "eyes", which connect to "verbal channel" and "visual channel" respectively. The arrows coming into "ears" and "eyes" are labeled "speech" and "images" respectively.

# The Magic Number $7 \pm 2$

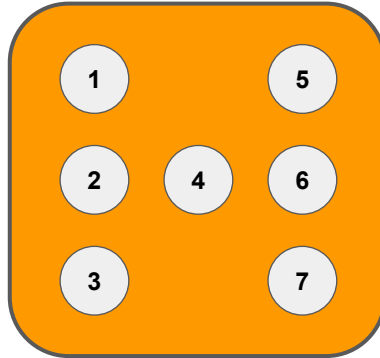


**short-term memory**

As long ago as the 1950s, researchers realized that the average human being can only hold a few items in short-term memory at a time. Initial estimates put the number at  $7 \pm 2$ ; more recent estimates put it as low as 4.

Description: a box labeled short-term memory that contains the numbers 1 through 7.

## The Magic Number $7 \pm 2$

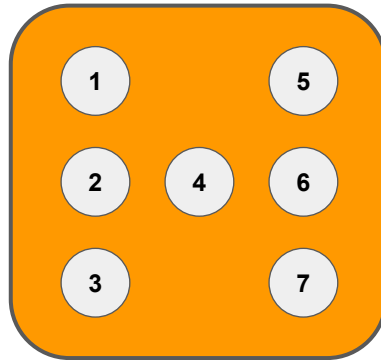


**short-term memory**

This is one of the reasons phone numbers are 6–8 digits long—it's hard to hold more digits than that in short-term memory long enough for the dial to go around.

Description: the short-term memory box from the previous slide containing the numbers 1 to 7, with a rotary telephone beside it.

# The Magic Number $7 \pm 2$

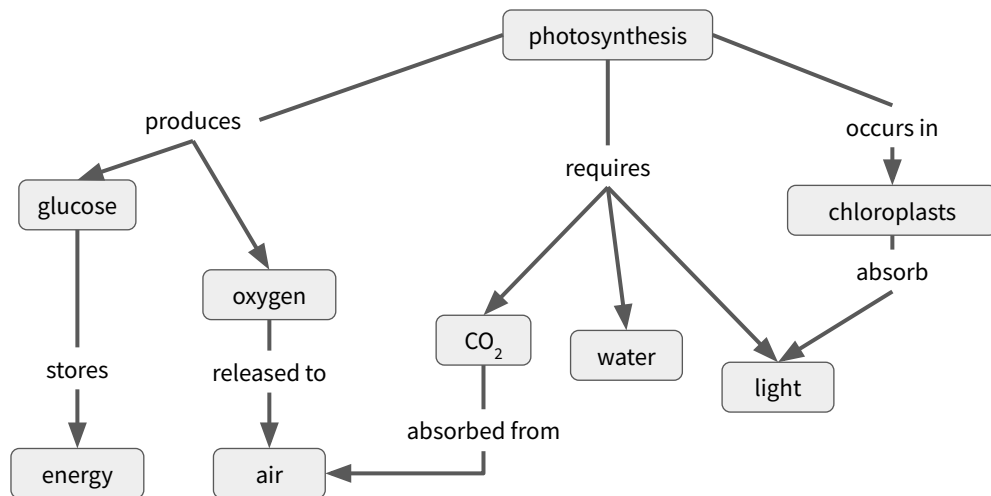


**short-term memory**

It may also be why the basic unit of human social organization is half a dozen people. From sports teams and military formations through knitting circles to meetings, that seems to be the limit of the number of people who can all be talking or working with one another at a time. Beyond that, larger groups quickly break down into sub-groups. (If you have 20 people in a meeting, you either have three meetings going on or one meeting of half a dozen people with a bunch of bystanders.)

Description: the short-term memory box from the previous slide containing the numbers 1 to 7, with a rotary telephone on its left, and a small crowd of people on its right.

## Count Them All

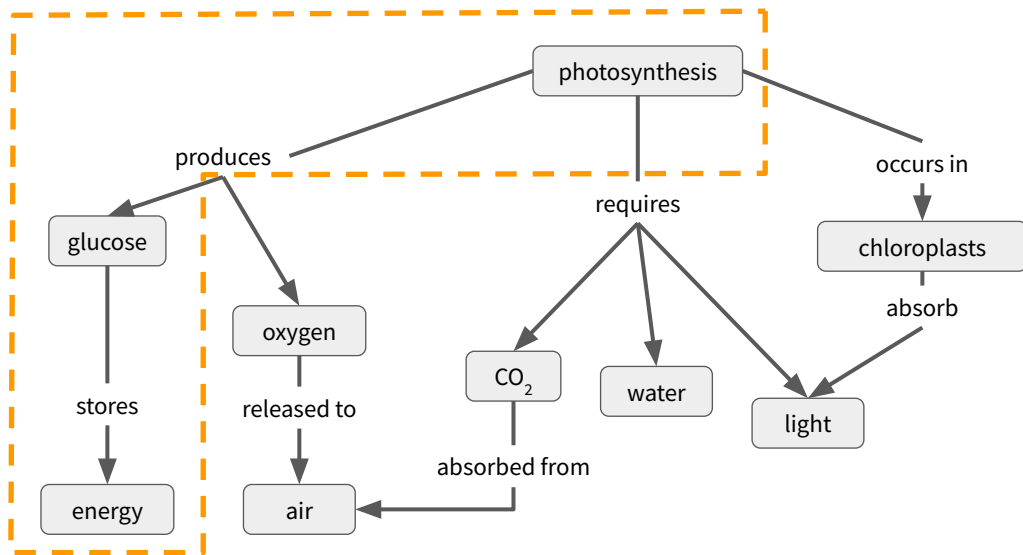


7

So let's return once more to our concept map of photosynthesis. There are nine nodes and seven labelled relationships (some of which are one-to-many, but we'll ignore that for now). That makes a total of sixteen things for a learner to absorb, and *that* tells us that this is too much material to tackle at once.

Description: concept map of photosynthesis including nine concepts, which are the nodes in the graph, and seven labeled arcs, some of which are one-to-many.

## Count Them All



8

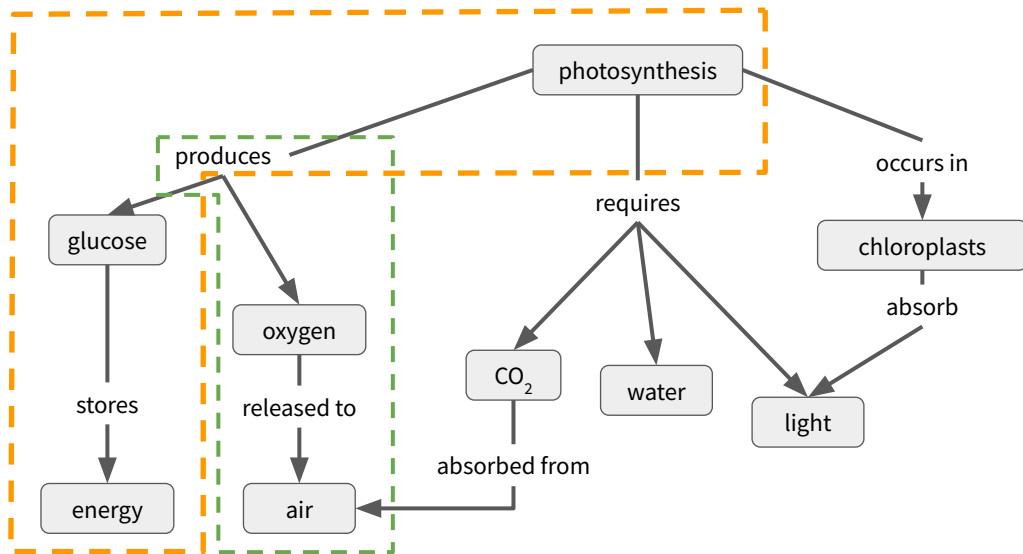
Here's one digestible chunk.

Description: the photosynthesis concept map with a dashed box drawn around a small subsection containing

- The top node labeled "photosynthesis"
- An arc from that node to the node labeled "glucose"
- An arc from the "glucose" node to the node labeled "energy". This arc is labeled "stores".



## Count Them All



9

And here's another. We're using the concept map to guide our choice of chunk size as well as subgoals.

The photosynthesis concept map shown before with the same highlighted section including photosynthesis, glucose, and energy, and a second highlight added that includes:

- The arc labeled “produces”
- Which connects to the node labeled “oxygen” as well as the node labeled “glucose”.
- An arc from the node labeled “oxygen” to the node labeled “air”. This arc is labeled “released to”.

*Hold on a second...*

10

But wait a second. If short-term memory can only hold half a dozen things at once, how are we able to think about bigger things? The answer is that if things frequently occur together, we store and recall them in *chunks*.

# Exercise

Memorize the symbol used to represent each number.

|   |                                                                                   |   |                                                                                   |   |                                                                                     |
|---|-----------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------|
| 1 |  | 4 |  | 7 |  |
| 2 |  | 5 |  | 8 |  |
| 3 |  | 6 |  | 9 |  |

11

Here's an example. Take a few seconds and memorize the symbol used to represent each digit.

Description: The numbers 1 to 9, each of which is beside a shape made of two, three, or four lines joined at right angles. There does not appear to be any pattern to the shapes.

*Let's try that again...*

Let's try that again...

# Exercise

Does this make it easier?

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

13

This is easier, isn't it? Our brains are very good at seeing patterns—so good, in fact, that they will see patterns that aren't there. As teachers, we can use this to reduce cognitive load and accelerate learning.

Description: the digits 1 to 9 arranged in a 3 by 3 grid similar to a telephone keypad. The lines between the numbers form the same right-angle shapes shown previously, but now that the numbers are arranged this way it is easy to see the grid pattern.

## Faded Examples

$$\begin{aligned}(4x + 8)/2 &= 5 \\ 4x + 8 &= 2 * 5 \\ 4x + 8 &= 10 \\ 4x &= 10 - 8 \\ 4x &= 2 \\ x &= 2/4 \\ x &= 1/2\end{aligned}$$

14

One way to do that is called *faded examples*. First, the instructor works through a problem step by step.

Description: seven lines of algebra showing how to reduce a simple equation to find 'x'.

## Control Group

$$\begin{aligned}(2x + 8)/4 &= 1 \\ x &= \_\_\end{aligned}$$

The control group is then asked to solve this problem immediately. They have to recall both the vocabulary of basic arithmetic and the “grammar” of the solution strategy.

Description: two lines of algebra, the first of which presents an equation with a single unknown ‘x’, and the second of which has “x equals” and a blank on the right for the solution.

## Treatment Group

$$\begin{aligned}
 (3x - 1) \cdot 3 &= 12 \\
 3x - 1 &= \underline{\hspace{1cm}} / \underline{\hspace{1cm}} \\
 3x - 1 &= 4 \\
 3x &= \underline{\hspace{1cm}} \\
 x &= \underline{\hspace{1cm}} / 3 \\
 x &= \underline{\hspace{1cm}}
 \end{aligned}$$

$$\begin{aligned}
 (5x + 1) \cdot 3 &= 4 \\
 5x + 1 &= \underline{\hspace{1cm}} \\
 5x &= \underline{\hspace{1cm}} \\
 x &= \underline{\hspace{1cm}}
 \end{aligned}$$

$$\begin{aligned}
 (2x + 8) / 4 &= 1 \\
 x &= \underline{\hspace{1cm}}
 \end{aligned}$$

The treatment group, on the other hand, works through three problems *in order*. Each gives them less *scaffolding*, which means they have to do more work, but it reinforces their understanding of the solution strategy. The treatment group does more work, but learns faster and better.

Description: three columns, each of which contains an equation with one unknown 'x'. The left column shows the same steps as the original 7-line problem, but with some blanks to be filled in. The middle column skips a few steps, so it shows fewer lines, but still has blanks to be filled in. The right column shows the same two lines shown earlier: the first line is an equation with an unknown 'x', and the second is 'x equals' and a blank to be filled in.



## Faded Examples

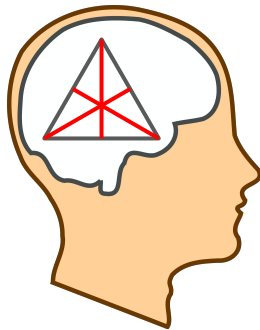
*Reduces germane load*

17

Faded examples reduce germane cognitive load by allowing learners to focus on the steps without simultaneously trying to have to pick a strategy.

## Faded Examples

$$(2x + 8)/4 = 1$$



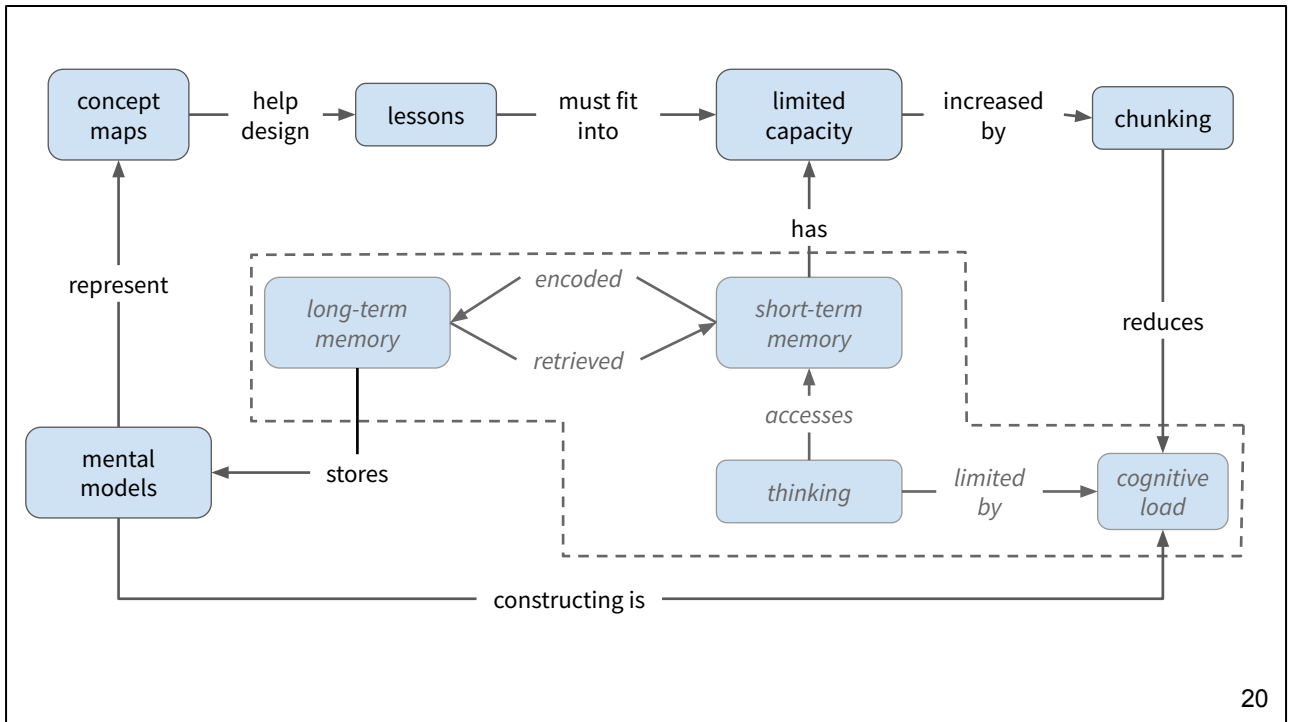
Faded examples also encourage learners to form the right chunks by drawing their attention to useful patterns.

# Exercise

1. Write short point-form instructions to do something you would teach.
2. Replace a few related pieces with \_\_'s or some other marker.
3. Swap with a partner.



Think of a small pattern you frequently use when solving problems. Create an example of this pattern in action, then fade some of the details. What are you drawing attention to in your choice of what to fade? Share your example with your neighbor and work through theirs.



Concept map for mental capacity.

Concept map for cognitive capacity.